OpenWebSearch.eu — Project Overview

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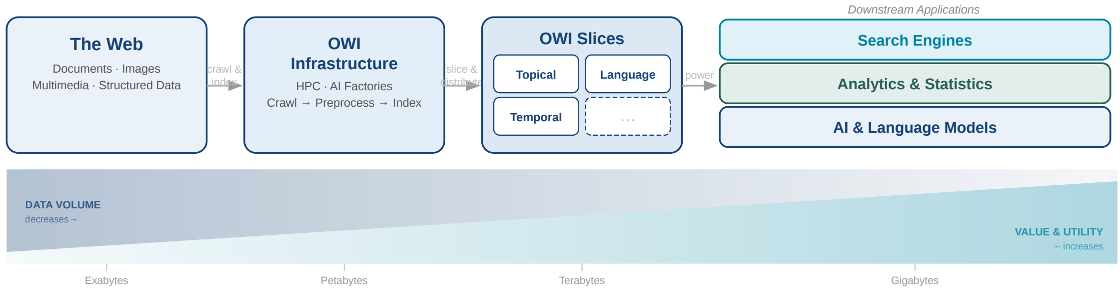
Overview of goals, results, impact, and outlook

Overall Goal

Build and pilot the core of a **European Open Web Index (OWI)** and a federated **Open Web Search and Analysis Infrastructure** at **TRL 5**.



Project proposition

Reduce the entry costs for European researchers, innovators, and organisations that want to use the Web for search, analytics, and AI.

Operational interpretation

Build not only a dataset, but a reusable stack for **crawling, enrichment,**

indexing, distribution, and retrieval.

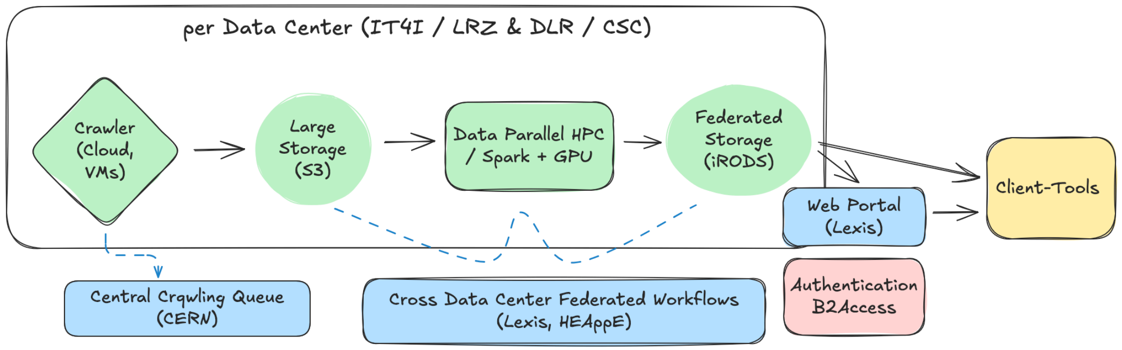
Project Objectives I

1. Open technology stack

- open pipelines for crawling, preprocessing, indexing, and access
- reusable components instead of one closed vertical service
- basis for search, analytics, and AI-oriented retrieval

2. Resource provision

- federated deployment across European HPC and data-centre infrastructure
- shared storage, orchestration, and publishing capabilities
- technical basis for future scaling via EuroHPC and AI Factory contexts



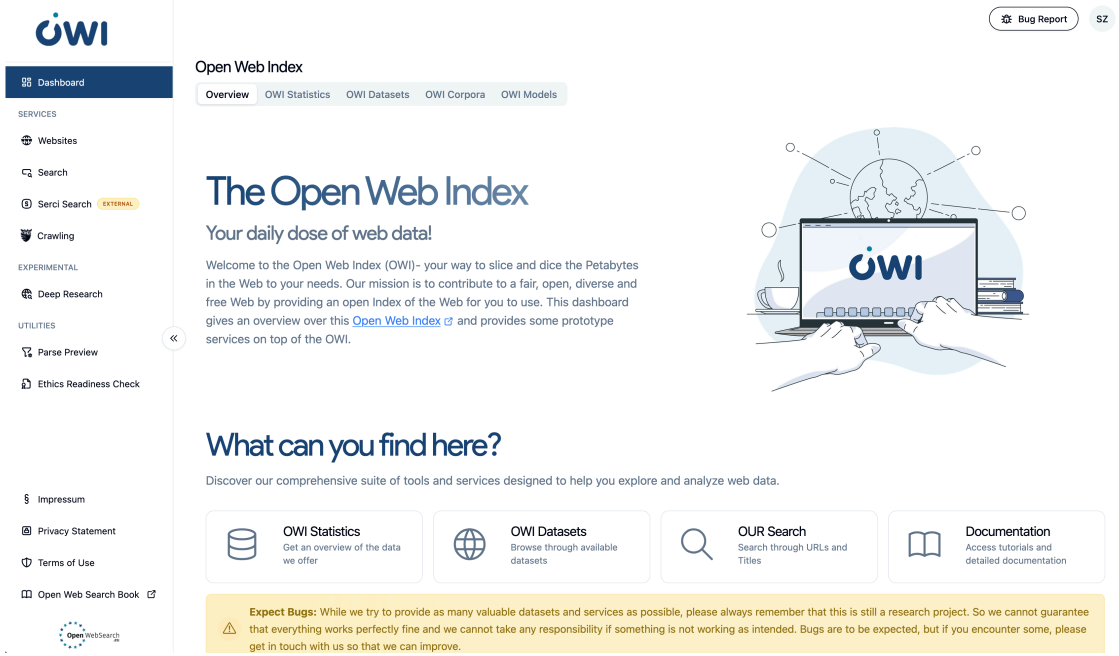
These two objectives define the infrastructural core of the project: without them, there is no operational OWI.

Project Objectives II

3. Added-value services
- dashboards, access tools, APIs, and prototype search services
- transparency and webmaster-facing services
- lower barriers for external use of the OWI

4. Ecosystem building
- applications, community programme, governance, and exploitation work
- legal and ethical framing for responsible use
- uptake beyond the original consortium

These objectives matter because an open index is only useful if it is accessible, governable, and used.



Core Result: OWI, Infrastructure, Community

Open Web Index
- **39.6B** fetched pages
- **12.6B** unique pages
- **10B+** indexed pages
- **500+** daily snapshots

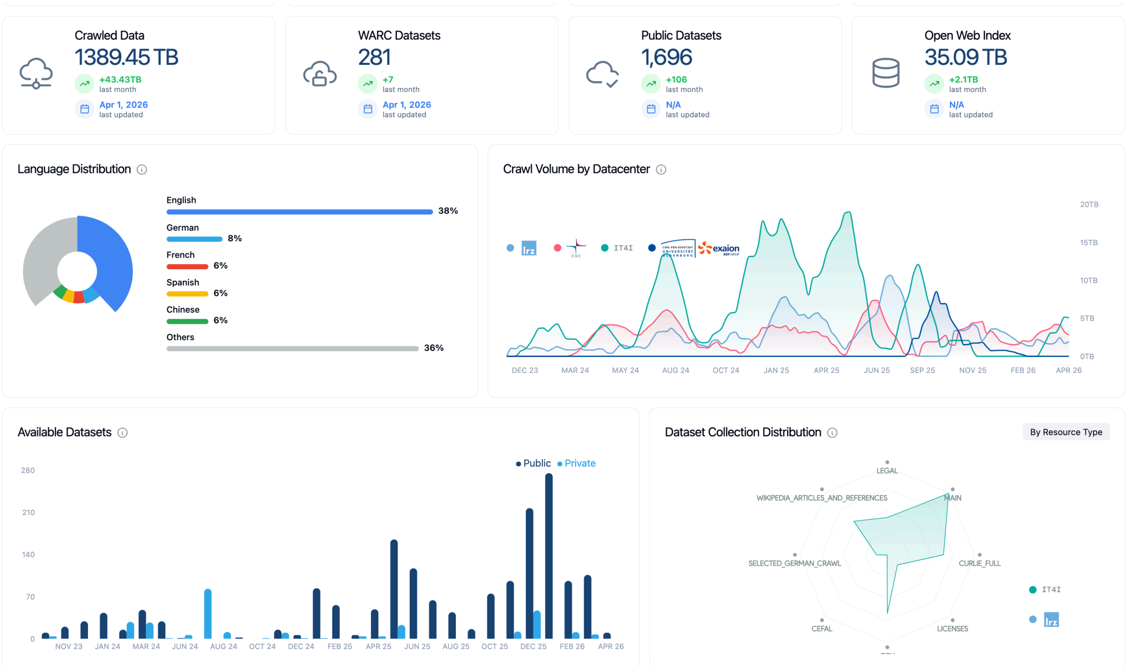
Detailed results: **WP1-WP3**

Federated Infrastructure
- **1.25 PiB** crawled raw data
- workflows across **3 EuroHPC JU centres + 3 further data centres**
- LEXIS, iRODS, S3, HEAppE as operational backbone

Detailed results: **WP5**, plus **WP1-WP3**

Community & Uptake
- **180+** community members
- **16** third-party projects funded
- governance and continuation work established

Detailed results: **WP6-WP8**



Research

Infrastructure

NGOs

Businesses

The OWI as the Technical Core

What was delivered

- **39.6B fetched pages**
- **12.6B unique pages**
- **1.25 PiB** crawled raw data
- **10B+ indexed pages**
- **75M domains** and **200M hosts**
- **500+ daily snapshots** in CIFF + Parquet
- **58 public OWIE datasets** with **119M embeddings**

Why this is the core result

- Europe now has a working, open, federated web-index backbone instead of only a concept
- the pipeline is operational end to end: **crawl -> preprocess -> enrich -> index -> distribute -> search**
- the index is not only downloadable; it is also **queryable and application-ready**
- the project demonstrated that open web search infrastructure can run across European HPC and cloud environments

> Most important technical message: an **Open Web Index at meaningful scale is feasible**.

Main Results by Work Package

WP1-WP2
- federated crawling
- aggregated crawl transparency
- webmaster console
- preprocessing, content analysis, embeddings

WP4
- Mosaic, MosaicRAG, EO search, argument search, recommender
- user-centred evaluation of applications

WP3-WP5
- CIFF + Parquet index production
- remote index, OWIlix, OURRS
- federated storage and workflow execution

WP6-WP8
- legal, ethical, and societal groundwork
- dissemination, exploitation, and community building
- coordination, FSTP, governance, and continuation planning

Work Package Overview

WP	Outcome at review level
WP1	Federated crawling, OWSI/OCSI aggregation, Open Webmaster Console, structured harvesting
WP2	Scalable preprocessing, content analysis, nutrition labels, geoparsing, embeddings
WP3	Index production, remote index, OWIlix, OURRS, evaluation collections
WP4	Search applications and user evaluation: Mosaic, MosaicRAG, EO search, argument search, recommender
WP5	Federated data infrastructure across European compute and storage sites
WP6	Legal, ethical, societal, governance, sustainability groundwork
WP7	Dissemination, exploitation, communication, community growth
WP8	Coordination, FSTP execution, policy engagement, post-project continuation setup

The technical WPs produced the backbone; the non-technical WPs made it usable, discussable, and continuable.

Impact

Strategic impact

- reduces dependency on proprietary, non-European web indices
- creates a shared resource for search, analytics, and AI experimentation
- establishes an operational model relevant for **AI Factories**, **EOSC**, and data spaces

Policy and ecosystem impact

- open web search entered discussions with the EC, EuroHPC, and the European Parliament
- three `#ossym` symposia and three WOWS workshops helped establish the field
- public launch of the OWI in June 2025
- governance work converged on a **MoU-based transition model**

Quantifiable signals

- host coverage target of **500M** exceeded with **516M hosts**
- **180+** community members
- **93 papers** published in RP2
- estimated **~EUR 4.5B** net economic and societal benefits over a decade

Realistic assessment

- strong proof of feasibility
- important services and assets now exist
- not yet a finished product or 24/7 service
- main remaining gaps are funding, operations, and legal-regulatory conditions

Sustainability: MoU-Based Continuation

Why an MoU

- the project created infrastructure, data products, and community assets that should not end with the funded pilot
- a permanent legal entity was assessed, but not yet realistic as an immediate transition step
- the **Memorandum of Understanding** is the most practical bridge between project end and longer-term institutionalisation

What the MoU enables

- coordinated continuation of core OWI operations on a best-effort basis
- shared governance and clear responsibilities among continuing partners
- continuity for community, infrastructure, and service access while follow-up structures mature

What it does not solve on its own

- no guaranteed base funding
- no automatic move to full production-grade 24/7 operations
- no resolution of the wider legal and regulatory barriers for commercial exploitation

Why it still matters

- it keeps the project legacy actionable rather than symbolic
- it creates a credible handover model for EuroHPC / AI Factory related uptake
- it gives time to move from pilot coordination to durable European governance

Follow-Up on Previous Review Recommendations

1. Ethical / legal implications of making the index available

- ethical self-assessment, takedown handling, transparency mechanisms, webmaster controls
- legal studies resulted in an **Open Web Index License** for scientific use
- policy recommendations were produced and communicated to EU policy processes

2. Sustainability and exploitation after project end

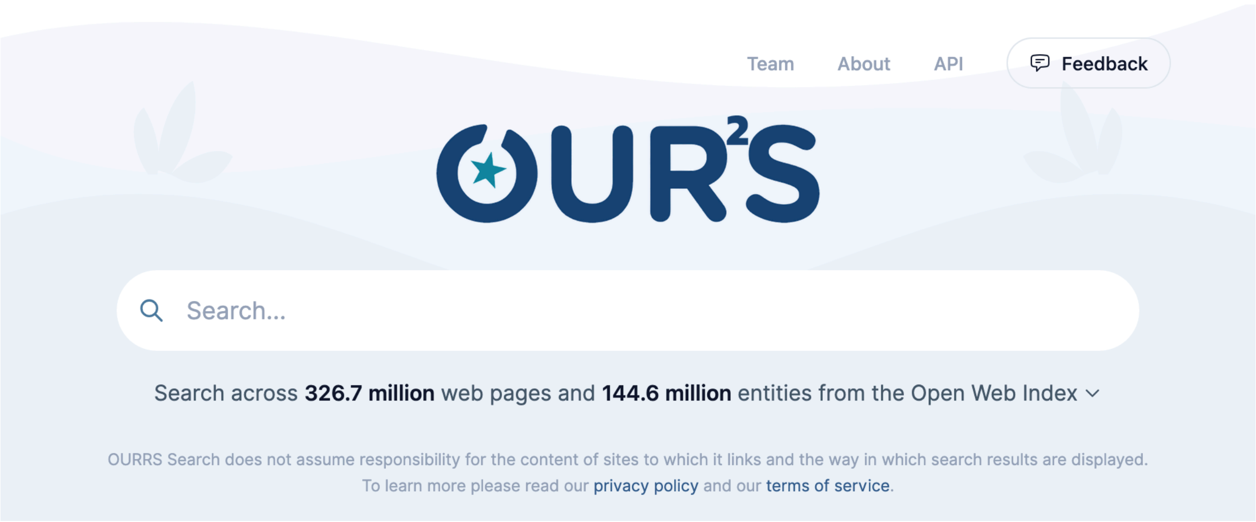
- Open Search Foundation coordination plus a **MoU-based best-effort continuation**
- governance options assessed; transitional structure defined
- follow-up projects secured, including **SOURCE** starting in **April 2026**
- AI Factory / AI Factory Antenna discussions now provide a realistic scaling path

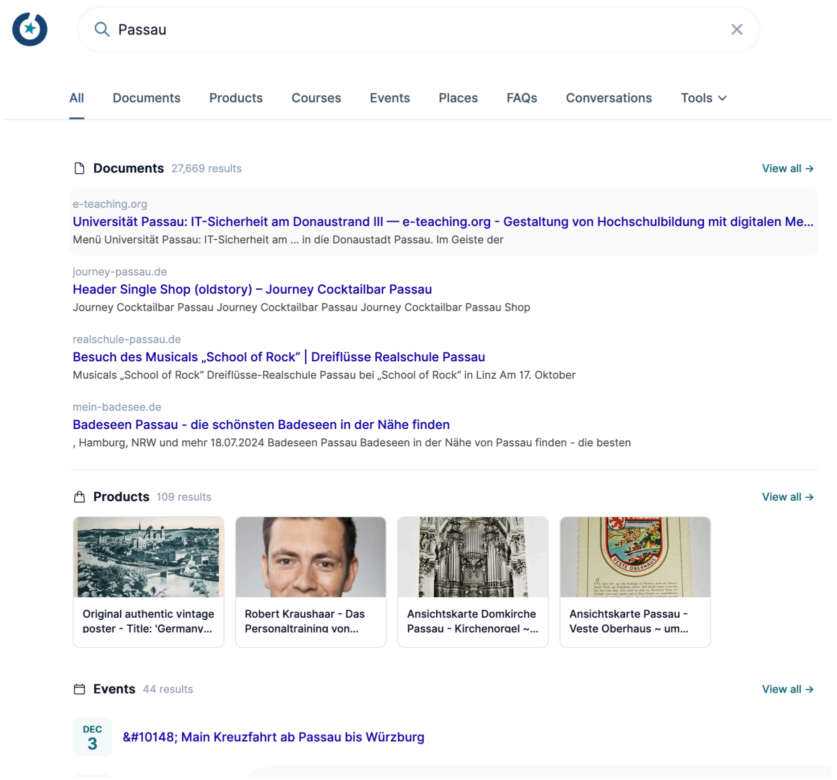
3. Adapting to LLM / GenAI developments

- Resilipipe GPU added large-scale embedding generation
- **German Commons** and **WebFAQ** created for European LLM work
- model and dataset hosting capabilities added to the platform
- **MOSAIC-RAG** and **OURRS** demonstrate practical RAG-oriented retrieval on top of OWI

Detailed results in **WP2**, **WP3**, **WP6**, **WP7**, and **WP8**.

Demo: OURRS as the User-Facing Proof Point





Why this demo matters

- shows OWI is not only a dataset archive
- provides a hosted search UI and API over OWI content
- lowers adoption barriers for application developers and RAG systems

Demo flow

1. issue a query against live OWI-backed search
2. inspect retrieval quality and result transparency
3. connect the demo back to the pipeline and access story

Claim to demonstrate: **Open infrastructure can support usable retrieval services, not just offline research artefacts.**

Final Takeaway

Delivered

- a federated European Open Web Index at meaningful scale
- the end-to-end open pipeline to build and refresh it
- access modes for datasets, remote querying, and live search
- applications, community, governance, and policy traction

What matters most after M42

- the project de-risked the core technical question
- the next challenge is no longer "can this be built?"
- it is now "how does Europe sustain and scale it?"

> OpenWebSearch.eu turned European open web search from an idea into operating infrastructure.